

Complex Analysis Homework #7

due 2026 MAY 11, 10:00 a.m.

1 Assignments

All problems are in the handout.

- I. Exercise D.163;
- II. Exercise D.172;
- III. Exercise D.182;
- IV. Exercise 1.82;
- V. Exercise 3.100;
- VI. Exercise 3.104;
- VII. Exercise 3.105;
- VIII. Exercise 3.110;
- IX. Exercise 3.111;
- X. Exercise 3.113;
- XI. Exercise 3.114;
- XII. Exercise 3.118;
- XIII. Exercise 3.121;
- XIV. Exercise 3.122*.

Each problem weighs 10 points, while Problems I, II and III weigh 5 points, Problems IV and XIV weigh 20 points. Problem XIV marked with “*” is an extra-edit problem. The full mark is 145 while the highest possible score is 160; see Section 3.

Caution: please make sure that your handwriting is recognizable, otherwise you only get partial credit for the recognizable part.

重要提醒: 证明时必须说明所用的定理定义等依据, 并且步骤详细, 否则不得分!

2 Tips

The four steps of math learning are:

- L1. understand the ideas, e.g. why the theorems hold,
- L2. memorize the packaging, i.e. the summary handout,
- L3. apply the math to solve problems,
- L4. integrate new learning into your knowledge system.

Doing homework is in the early stage of L3, so it is often time-efficient to do step L2 before attempting the homework problems.

Ask yourself two questions after finishing a problem:

- “What previous knowledge have I used?”
- “What is the connection between the previous knowledge to the new concepts and theorems?”

3 Extra credits

Additional 10% credits will be given to you if you typeset your solutions in $\text{\LaTeX}$. You can also get partial extra credit for typesetting solutions of *some* problems.